Transport: for a type family P over A and a path p from x to y there is a function from P(x) to P(y)

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled lem:transport (source lines 761-764); the proof span 766-774 is the author's first proof environment following it, retained line for line.

Editorial note. The author's second proof at the following lines is not counted and not retained on the card face.

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: transport along a path in a type family. Only the counted lemma and its first author proof are retained; the author's second proof and the surrounding commentary on dependent functions remain uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent type families P over A and the identity type of A with its reflexivity element.
- The induction principle for identity types (path induction) with its conversion rule.
- The identity function on a type and dependent product types.

Lemma 0.0.1 (Transport)**.** *Suppose that P is a type family over A and that $p : x =_A y$. Then there is a function $p_* : P(x) \to P(y)$.*

First proof. Let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family defined by

$$D(x, y, p) :\equiv P(x) \to P(y).$$

Then we have the function

$$d :\equiv \lambda x.\, \mathsf{id}_{P(x)} : \prod_{x:A} D(x, x, \mathsf{refl}_x),$$

so that the induction principle gives us $\mathsf{ind}_{=_A}(D, d, x, y, p) : P(x) \to P(y)$ for $p : x = y$, which we define to be p_*. □